

Case study practical

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Laptop Not Powering On – 5 Troubleshooting Steps:-

Step 1: Check the Power Socket

Check whether the wall socket is working properly. Try another socket if needed.

Step 2: Check the Charger

Inspect the charger, cable, and connector for cuts, damage, or loose connections. Check whether the charging indicator turns on.

Step 3: Check the Charging Port

Inspect the laptop charging port for dust, damage, or looseness. Make sure the charger fits properly.

Step 4: Perform a Power Reset

Unplug the charger, remove the battery if removable, and hold the **Power button for 15–20 seconds**. Reconnect the charger and try powering on.

Step 5: Check Battery/Hardware

If the laptop still does not start, check whether the battery is damaged or completely dead. If the charger and battery are fine, the problem may be with the **motherboard or power circuit**.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

 Hint: Consider checking both hardware connections and basic display settings.

Blank Monitor – Short Troubleshooting Steps:-

1. **Check power** – Ensure the monitor is switched on and power LED is working.
2. **Check cable** – Reconnect the HDMI/DisplayPort/VGA cable properly.
3. **Check input source** – Select the correct HDMI/DP/VGA input on the monitor.
4. **Check GPU/RAM** – Reseat the GPU and RAM and check their connections.
5. **Test another monitor** – Connect another working monitor to identify whether the problem is with the monitor or PC.
6. **Check display settings** – Use **Win + Ctrl + Shift + B** to reset the graphics driver.

7. **Confirm fix** – Restart and check that the display works normally.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Wireless Mouse Not Working – Troubleshooting:-

1. **Check the Mouse** – Turn the mouse ON, replace the battery, and check if the sensor light works.
2. **Check USB Receiver** – Remove and reconnect the wireless receiver properly.
3. **Try Another USB Port** – Connect the receiver to a different USB port. If it works, the original port may have a problem.
4. **Test Another Mouse** – Connect a different working wireless/wired mouse. If it works, the original mouse may be faulty.
5. **Check System Settings** – Open **Device Manager** → **Mice and other pointing devices** and check for driver errors. Update or reinstall the mouse driver if needed.
6. **Confirm the Cause** – Test the mouse on another computer. If it works there, the issue is likely with the original computer's USB port, driver, or system settings.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

Gaming PC Overheating – 3 Hardware Causes:-

1. Faulty or Slow Fans

- **Check:** See whether the CPU/GPU/case fans are spinning properly.
- **Fix:** Clean or replace the faulty fan.

2. Dust Buildup

- **Check:** Open the case and inspect the fans, heatsink, and air vents for dust.
- **Fix:** Clean the dust using compressed air.

3. Dried Thermal Paste

- **Check:** Inspect the thermal paste between the CPU and heatsink.
- **Fix:** Remove the old paste and apply fresh thermal paste.